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10 / 10 submitted

Q1. Looping statements

CODE

```
#include<stdio.h>
int main()
{
    int n,rem,temp,sum=0;
    scanf("%d",&temp);
    while(temp!=0)
    {
        rem=temp%10;
        sum=sum*10+rem;
        temp=temp/10;
    }
    printf("reverse the digits=%d",sum);
    return 0;
}
```

INPUT

123

OUTPUT

reverse the digits=321

Q2. Looping statements

CODE

```
#include<stdio.h>
int main()
{
    int n,rem,temp,sum=0;
    scanf("%d",&temp);
    n=temp;
    while(temp!=0)
    {
        rem=temp%10;
        sum=sum*10+rem;
        temp=temp/10;
    }
    if(sum==n)
        printf("palindrome",sum);
    else
        printf("not a palindrome",sum);
    return 0;
}
```

INPUT

121

OUTPUT

palindrome

Q3. Looping statements

CODE

```
#include<stdio.h>
int main()
{
    int n,temp,rem,sum=0;
    scanf("%d",&temp);
    n=temp;;
    while(temp!=0)
    {
        rem=temp%10;
        sum=sum+(rem*rem*rem);
        temp=temp/10;
    }
    if(sum==n)
        printf("armstrong number",sum);
    else
        printf("not a armstrong number",sum);
    return 0;
}
```

INPUT

153

OUTPUT

armstrong number

Q4. Looping statements

CODE

```
#include<stdio.h>
int main()
{
    int i,j,n,isprime,sum=0;
    scanf("%d",&n);
    for(i=2;i≤n;i++)
    {
        isprime=1;
        for(j=2;j*j≤i;j++)
        {
            if(i%j==0)
            {
                isprime=0;
                break;
            }
        }
        if(isprime)
        {
            sum+=i;
        }
    }
    printf("%d",sum);
    return 0;
}
```

INPUT

10

OUTPUT

17

Q5. Looping statements

CODE

```
#include<stdio.h>
int main()
{
    int i,j,n;
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        for(j=1;j<=n;j++)
        {
            if(i==1||i==n||j==1||j==n)
                printf("*");
            else
                printf(" ");
        }
        printf("\n");
    }
    return 0;
}
```

INPUT

5

OUTPUT

```
*****
*   *
*   *
*   *
*****
```

Q6. Looping statements

CODE

```
#include<stdio.h>
int main()
{
    int n,i,j;
    scanf("%d",&n);
    for(i=n;i>=1;i--)
    {
        for(j=1;j<=i;j++)
            printf("* ");
        printf("\n");
    }
    return 0;
}
```

INPUT

OUTPUT

Q7. Looping statements

CODE

```
#include<stdio.h>
int main()
{
    int n,i,j;
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        for(j=1;j<=n-i;j++)
            printf(" ");
        for(j=1;j<=2*i-1;j++)
            printf("* ");
        printf("\n");
    }
    return 0;
}
```

INPUT

5

OUTPUT

```
      *
     * * *
    * * * * *
   * * * * * *
  * * * * * * *
 * * * * * * * *
```

Q8. Looping statements

CODE

```
#include<stdio.h>
int main()
{
    int n,i,j;
    scanf("%d",&n);
    for(i=1;i≤n;i++)
    {
        for(j=1;j≤n-i;j++)
            printf(" ");
        for(j=1;j≤2*i-1;j++)
            printf("* ");
        printf("\n");
    }
    for(i=n-1;i≥1;i--)
    {
        for(j=1;j≤n-i;j++)
            printf(" ");
        for(j=1;j≤2*i-1;j++)
            printf("* ");
        printf("\n");
    }
    return 0;
}
```

INPUT

4

OUTPUT

```

  *
 * * *
* * * * *
 * * * * *
  * * *
    *
```


Q9. Looping statements

CODE

```
#include<stdio.h>
int main()
{
    int i,j,n,a=1;
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        for(j=1;j<=i;j++)
            printf("%d ",a++);
        printf("\n");
    }
    return 0;
}
```

INPUT

5

OUTPUT

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

Q10. Looping statements

CODE

```
#include<stdio.h>
int main()
{
    int i,j,n,coeff=1;
    scanf("%d",&n);
    for(i=0;i<=n;i++)
    {
        for(j=0;j<=n-i;j++)
            printf(" ");
        for(j=0;j<=i;j++)
        {
            if(i==0||j==0)
                coeff=1;
            else
                coeff=coeff*(i-j+1)/j;
            printf("%4d",coeff);
        }
        printf("\n");
    }
    return 0;
}
```

INPUT

5

OUTPUT

```
      1
     1 1
    1 2 1
   1 3 3 1
  1 4 6 4 1
 1 5 10 10 5 1
```